# ARJUNA JEE 2027

Lecture -01 | Mathematics

# BINOMIAL THEOREM

By- Sachin Jakhar Sir



# Topics *to be covered*

A **Derangement**

B **Pascal Triangle**

C **Binomial Expansion**

D **Question Practice**



# H.W. Discussion

DIBY-X

Find number of integral solutions of following equation :

$xyz = 2^8 \cdot 3^5 \cdot 5^4$, $(x, y, z \in N)$

Ans: 14175

## Question [JEE Mains-2023]

[Ans. : (D)]

The number of triplets $(x, y, z)$ where $x, y, z$ are distinct non negative integers satisfying $x + y + z = 15$, is

(A) 136

(B) 80

(C) 92

(D) 114

(W)

distinct → अलग - अलग.

$x + y + z = 15$

$^{15+3-1}C_{3-1} = {}^{17}C_2$

C-I

$x = y = z$ → 1.

C-II

$x = y \neq z$

$x \neq y = z$

$x = z \neq y$

→ $7 \times 3 = 21$.

$x, y, z$ distinct (Required)

C-I : $x = y = z = 5$ — (1) ✓

$3x = 15 \Rightarrow x = 5$

C-II : $x = y \neq z$

$2x + z = 15$

| | | |
|---|---|---|
| 0 | 15 | 4 7 |
| 1 | 13 | 5 5 X |
| 2 | 11 | 6 3 |
| 3 | 9 | 7 1 |

$\text{Inf} = {}^{17}C_2 - (21 + 1)$ ✓

## Question [JEE Mains-2023]



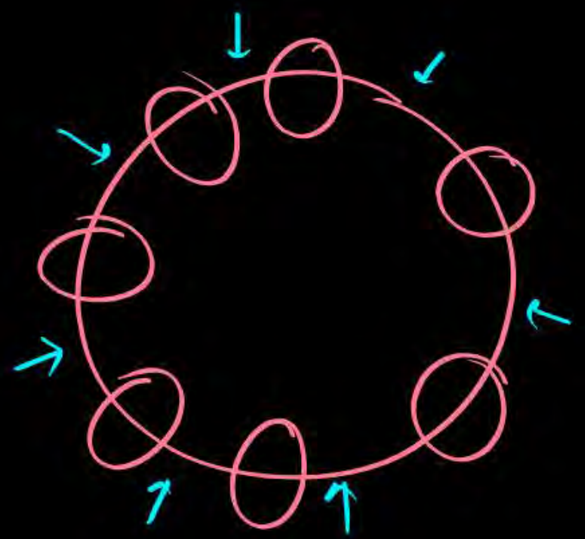
The number of ways, in which 5 girls and 7 boys can be seated at a round table so that no two girls sit together, is

A 720

B $126(5!)^2$

C $7(360)^2$

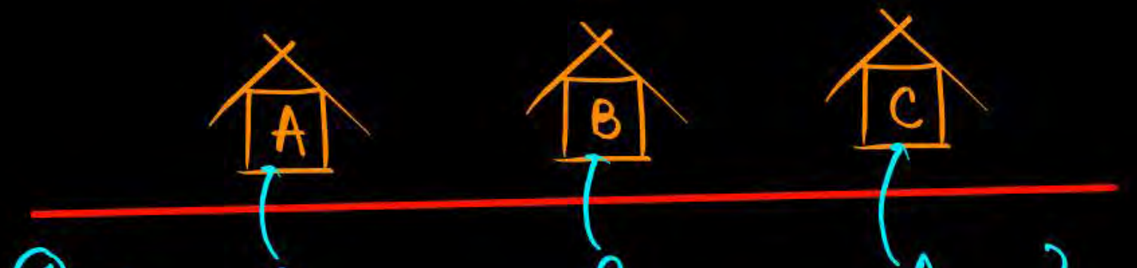
D $7(720)^2$



$6! \times {}^7C_5 \times 5!$

$6 \times 5! \times 21 \times 5!$

$126 \times (5!)^2$




① — B C A

① — C A B

2 ways.

# DE-ARRANGEMENT



| A | B | C | D |
|---|---|---|---|
| B | C | D | A |
| B | D | A | C |
| B | A | D | C |
| C | A | D | B |
| C | D | A | B |
| C | D | B | A |
| D | A | B | C |
| D | C | A | B |
| D | C | B | A |

*# Jitni "log" utni hi unki places hongi, koi bhi apni sahi jagah pr na baithe*

**# Number of ways =** 9 ways.

# DERANGEMENT

**# Tag Line:** ***Koi bhi apni shi jagah pr na baithe.***

**# Number of ways of Derangement of $n$-things = $D_n$**

$$D_n = (n!).\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots + \frac{(-1)^n}{n!}\right)$$

## FORMULA

$$D_n = (n!).\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots + \frac{(-1)^n}{n!}\right)$$

$n = 4$

$$D_4 = 4!\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right)$$

$$= 24\left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right) = 12 - 4 + 1 = 9$$

$n = 5$

$$D_5 = 5!\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right)$$

$$= 120\left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120}\right)$$

$$= 60 - 20 + 5 - 1$$

$$= 40 + 4$$

$$= 44$$

## REMEMBER AS RESULT



# Number of ways of deranging 1 thing = $D_1 = 0$

# Number of ways of deranging 2 things = $D_2 = 1$

# Number of ways of deranging 3 things = $D_3 = 2$

# Number of ways of deranging 4 things = $D_4 = 9$

# Number of ways of deranging 5 things = $D_5 = 44$

# Number of ways of deranging 6 things = $D_6 = 265$

## Bumper Question



Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card . Find number of ways such that:

(i) All cards are in correct envelopes = (1)

(ii) No card is placed in its corresponding envelope = $D_6 = 265$.

(iii) Exactly 3 cards are placed in correct envelopes = $^6C_3 \times 1 \times D_3$

$E_1$ $E_3$ $E_6$
$L_1$ $L_3$ $L_6$.

(iv) Exactly 4 cards wrongly placed = $^6C_4 \times D_4 \times 1$.

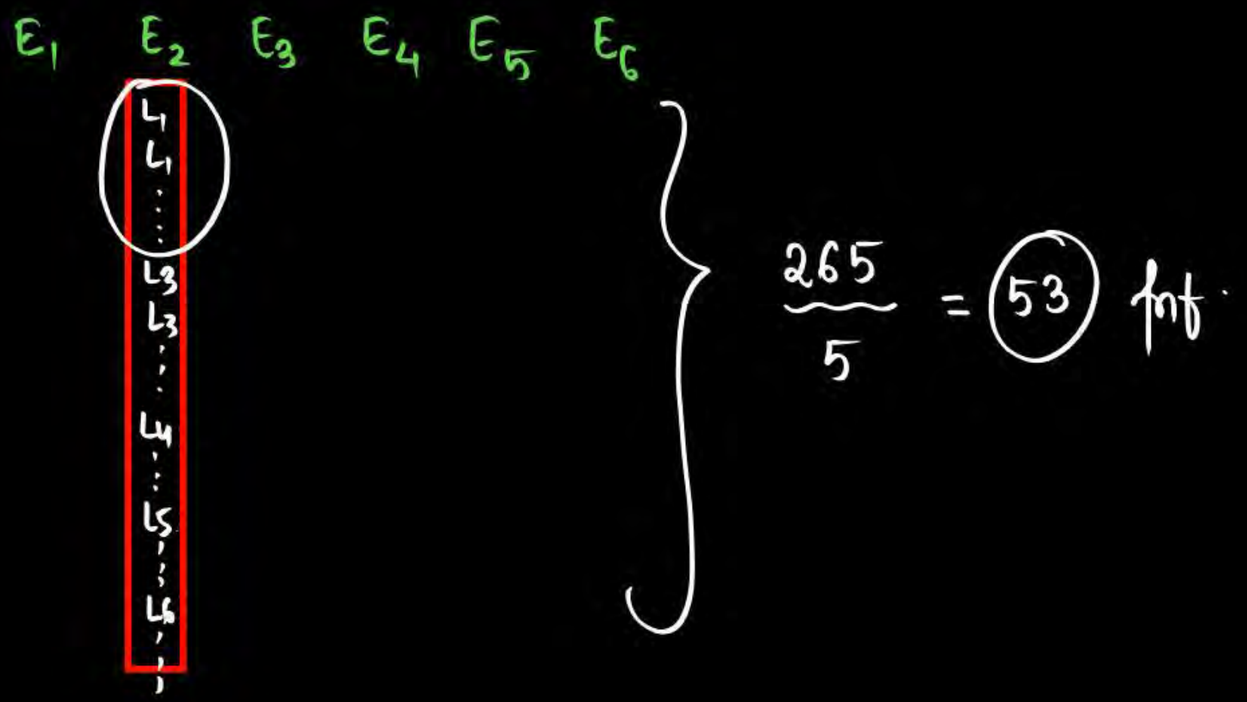
$E_1$ $\check{E}_2$ $E_3$ $\check{E}_4$ $\check{E}_5$ $E_6$
$L_1$ $L_2$ $L_3$ $L_4$ $L_5$ $L_6$

## Question [JEE (Adv.)-2014, Paper-2]



Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card . Find number of ways such that:

If no card is placed in the envelope bearing same number and moreover the card 1 is always placed in envelope numbered 2.



## Acidic Question



The number of words that can be formed by rearranging the letters of the word 'COVID' such that no letter is in it's own place and word should not start with a vowel is _____.





# PRINCIPLE OF INCLUSION & EXCLUSION



$$\# n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\# n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

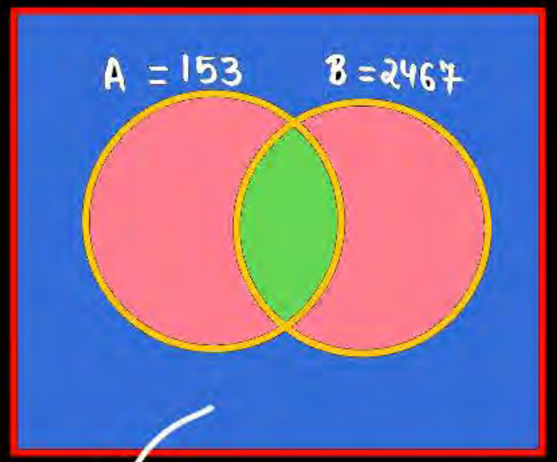
The number of permutations of the digits 1, 2, 3, .... 7 without repetition, which neither contain the string 153 nor the string 2467, is

ex :-

1234567 ✓

2467135 ✗

2641537 ✗




Total = $7!_0$

(A) → 153, 2, 4, 6, 7 → $5!_0$

(B) → 1, 5, 3, 2467 → $4!_0$

$A \cap B$ → 153, 2467 → $2!_0$

$n(A \cup B) = 5!_0 + 4!_0 - 2!_0$

$fnf = 7!_0 - (5!_0 + 4!_0 - 2!_0)$

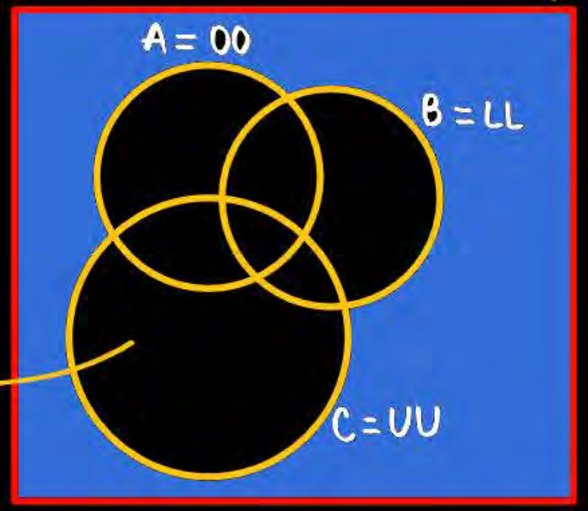
## Acidic Question

$$fnf = \frac{8!}{2!2!2!} - (n(A \cup B \cup C))$$

**How many different words can be formed using all the letters of the words HONOLULU if (no two alike letters are together)**

ONOHULUL

alike letters → OO, LL, UU.




$\frac{8!}{2!2!2!}$ → OO, L, L, U, U, H, N

A → $\frac{7!}{2!2!}$

B, C

A ∩ B → OO, LL, U, U, H, N → $\frac{6!}{2!}$

B ∩ C, C ∩ A

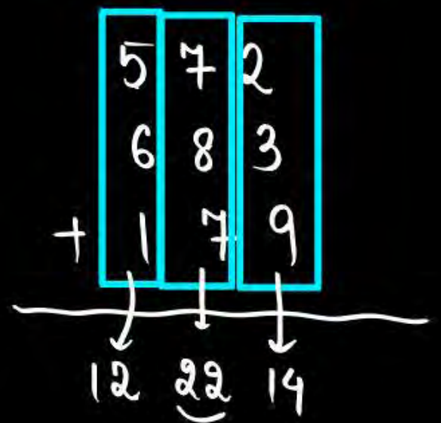
$$n(A \cup B \cup C) = 3 \times \frac{7!}{2!2!} - 3 \times \frac{6!}{2!} + 5!$$

A ∩ B ∩ C → OO, LL, UU, H, N → $5!$

# SUM OF ALL THE NUMBERS FORMED

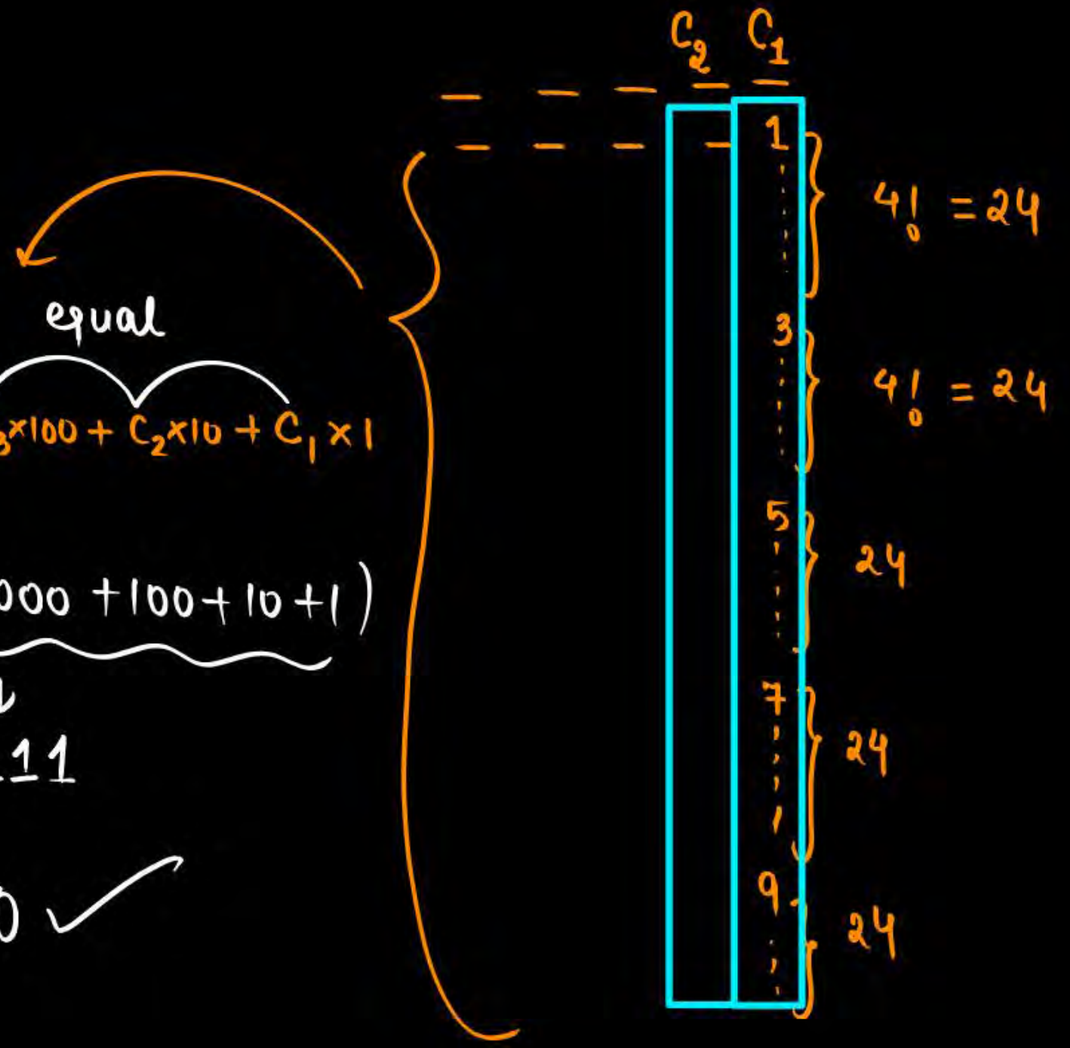


$$\begin{array}{r} {\scriptstyle 2\ \ 1} \\ 5\ 7\ 2 \\ 6\ 8\ 3 \\ +\ 1\ 7\ 9 \\ \hline 1\ 4\ 3\ 4 \end{array}$$




$= 12\times100 + 22\times10 + 14\times1 = 1434$

Ques:- ~~1~~, 3, 5, 7, 9 → RNA, 5 digit की Total no. → $5! = 120$ ✓




Sum = equal

$C_5 \times 10000 + C_4 \times 1000 + C_3 \times 100 + C_2 \times 10 + C_1 \times 1$

$= C_1 (10000 + 1000 + 100 + 10 + 1)$

$= 600 \times 11111$

$= 6666600$ ✓

Sum

$C_1 = 1 \times 24 + 3 \times 24 + 5 \times 24 + 7 \times 24 + 9 \times 24.$

$= 24 (1 + 3 + 5 + 7 + 9) = 600.$

Ques:- 1, 3, 5, 7, 9 → RNA, 5 digit के Total no. → $5! = 120$ ✓

If digits are in A.P.

$S = 13579 + _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ + 97531$

$S = 97531 + _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ + 13579$

---

$$S = \frac{111110 \times \overset{60}{\cancel{120}}}{\cancel{2}} = 6666600$$

## Acidic Question



Find sum of all 5 digit numbers that can be formed with help of 0, 1, 2, 6, 7 in which repetition of digits is not allowed.

∴ Hw.

## Question [JEE Mains-2023]



Sum of all the four digits numbers that can be formed using all the digits 2, 1, 2, 3 is_____.



# BINOMIAL THEOREM

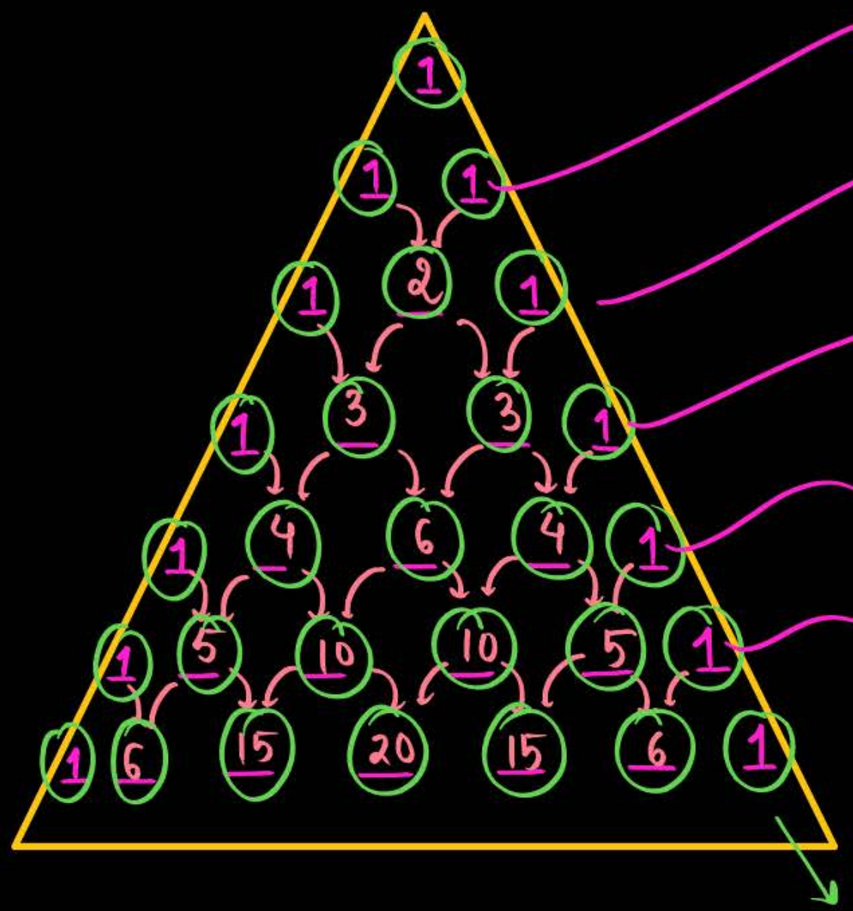
IInd Most imp. Chapter of class XI.

easy & formula based

2 Ques. – Mains
1 Ques. – Adv.

# PASCAL's TRIANGLE





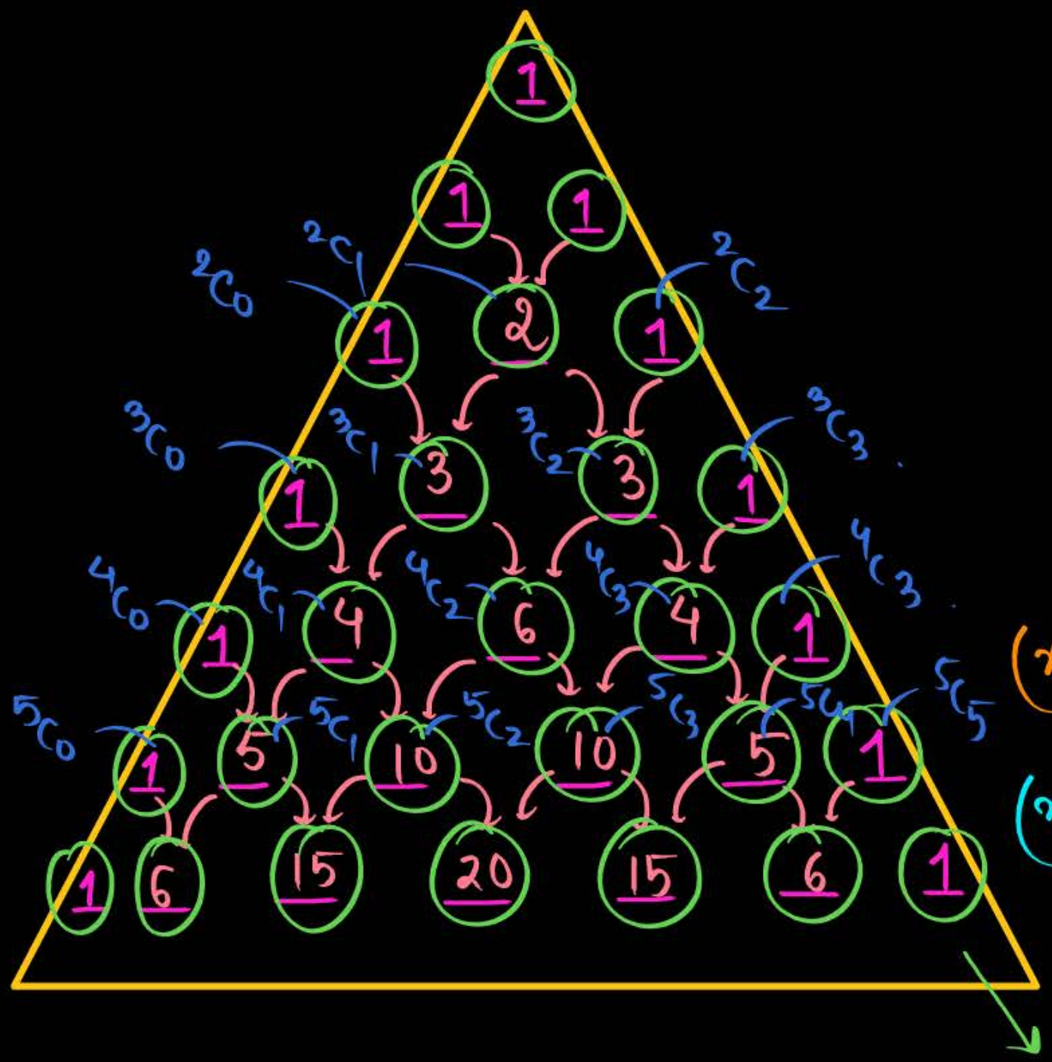

$(x+y)^1 = 1x + 1y$

$(x+y)^2 = 1x^2 + 2xy + 1y^2$

$(x+y)^3 = 1x^3 + 3x^2y + 3xy^2 + 1y^3$

$(x+y)^4 = 1x^4y^0 + 4x^3y^1 + 6x^2y^2 + 4x^1y^3 + 1x^0y^4$

$(x+y)^5 = 1x^5y^0 + 5x^4y^1 + 10x^3y^2 + 10x^2y^3 + 5x^1y^4 + 1x^0y^5$




$$(x+y)^1 = 1x + 1y$$

$$(x+y)^2 = 1x^2 + 2xy + 1y^2$$

$$(x+y)^3 = 1x^3 + 3x^2y + 3xy^2 + 1y^3$$

$$(x+y)^4 = 1x^4y^0 + 4x^3y^1 + 6x^2y^2 + 4x^1y^3 + 1x^0y^4$$

$$(x+y)^5 = \underset{{}^5C_0}{1}x^5y^0 + \underset{{}^5C_1}{5}x^4y^1 + \underset{{}^5C_2}{10}x^3y^2 + \underset{{}^5C_3}{10}x^2y^3 + \underset{{}^5C_4}{5}x^1y^4 + \underset{{}^5C_5}{1}x^0y^5$$

Bi - Nomial

Bi → Two

Nomial → Variable.

$2x + x$

$3x$

→ Monomial.

$x + y$

$x^2 + y^2$

$2x - 3y$

$a + \frac{1}{x}$

⋮

$x + y + z$

# NEWTON'S BINOMIAL EXPANSION



$T_1 \quad T_2 \quad T_3 \quad T_4 \quad \dots \quad T_{n+1}$

$$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1}b^1 + {}^nC_2 a^{n-2}b^2 + {}^nC_3 a^{n-3}b^3 \dots\dots + {}^nC_n b^n$$

$n \in N$

ex:- $$(a+b)^7 = {}^7C_0 a^7 b^0 + {}^7C_1 a^6 b^1 + {}^7C_2 a^5 b^2 - - - - + {}^7C_7 a^0 b^7$$

ex:- $(a+b)^7 = {}^7C_0 a^7 b^0 + {}^7C_1 a^6 b^1 + {}^7C_2 a^5 b^2 - - - - + {}^7C_7 a^0 b^7$

ex:- $(a-b)^7 = {}^7C_0 a^7 - {}^7C_1 a^6 b^1 + {}^7C_2 a^5 b^2 - - - - - {}^7C_7 a^0 b^7$

$(a + (-b))^7$

I II

## Example

**Write Binomial Expansion of $(2x-3y)^4$**

$(a+b)^4$ where $a = 2x$, $b = -3y$

$$(2x-3y)^4 = {}^4C_0(2x)^4 + {}^4C_1(2x)^3(-3y)^1 + {}^4C_2(2x)^2(-3y)^2 + {}^4C_3(2x)^1(-3y)^3 + {}^4C_4(-3y)^4.$$

$T_1$ $T_2$ $T_3$ $T_4$ $T_5$

$$(1) = (2-3)^4 = {}^4C_0 2^4 - {}^4C_1 2^3 \cdot 3^1 + {}^4C_2 2^2 3^2 - {}^4C_3 2^1 3^3 + {}^4C_4 3^4$$

# IMPORTANT NOTE

$$(a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + {}^nC_3 a^{n-3} b^3 + \ldots + {}^nC_n a^0 b^n$$

## # *Important Observations:*

(i) Sum of net power of '$a$' and '$b$' in each term is $n$

(ii) Number of terms in expansion = (power + 1) = $n+1$.

(iii) Compact form:

$$(a+b)^n = \sum_{r=0}^{n} {}^nC_r \, a^{n-r} b^r$$

(iv) General term:

$$T_{r+1} = {}^nC_r \, a^{n-r} b^r$$

$\left(3x^2 - \frac{4}{y}\right)^{11}$ → $7^{th}$ Term ??

Binomial coeff.

$$T_7 = T_{6+1} = {}^{11}C_6 \left(3x^2\right)^5 \left(\frac{-4}{y}\right)^6$$

$$T_7 = {}^{11}C_6\, 3^5 \cdot 4^6 \cdot \frac{x^{10}}{y^6}$$

coefficient · B.C.



# IMPORTANT NOTE



(v) $k^{th}$ term from last = $(n - k + 2)^{th}$ term from starting.

Next class.

(vi) Difference between coefficient and binomial coefficient.

(vii) To find "sum of all coefficients" put all parameters equal to 1.

*It is never about end result*

*It's always about Journey*

# Thank You

#futureIITians